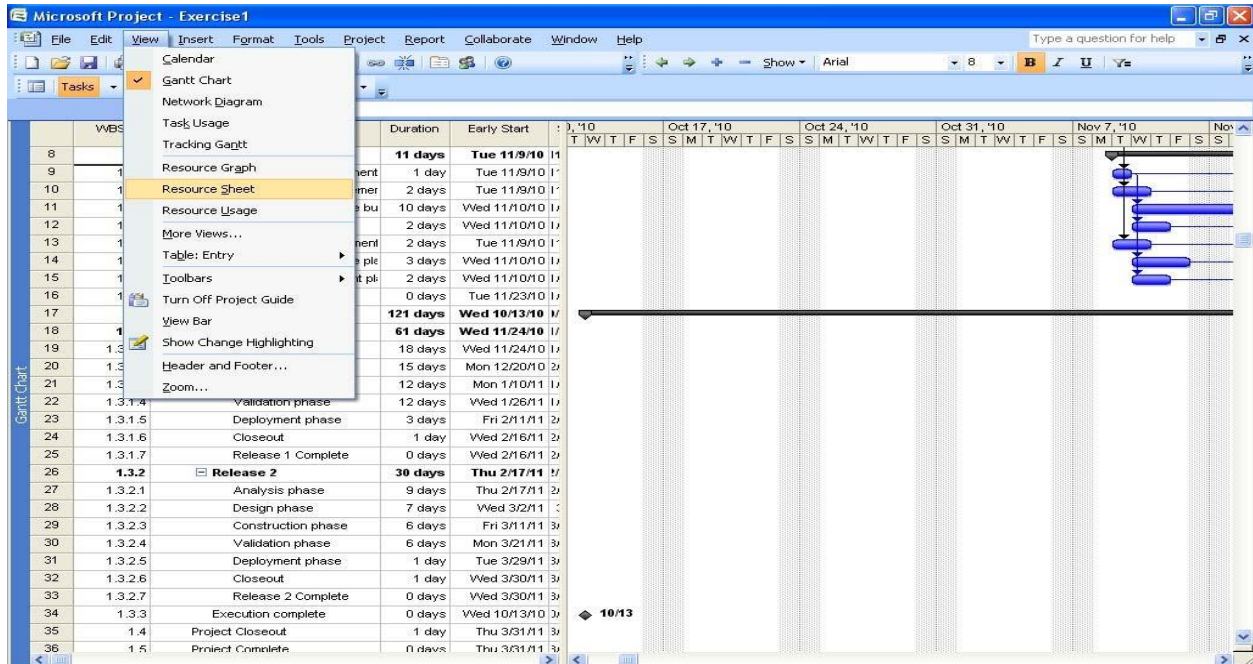


ADDING RESOURCES TO THE WBS

2.1 Displaying the Resource Sheet

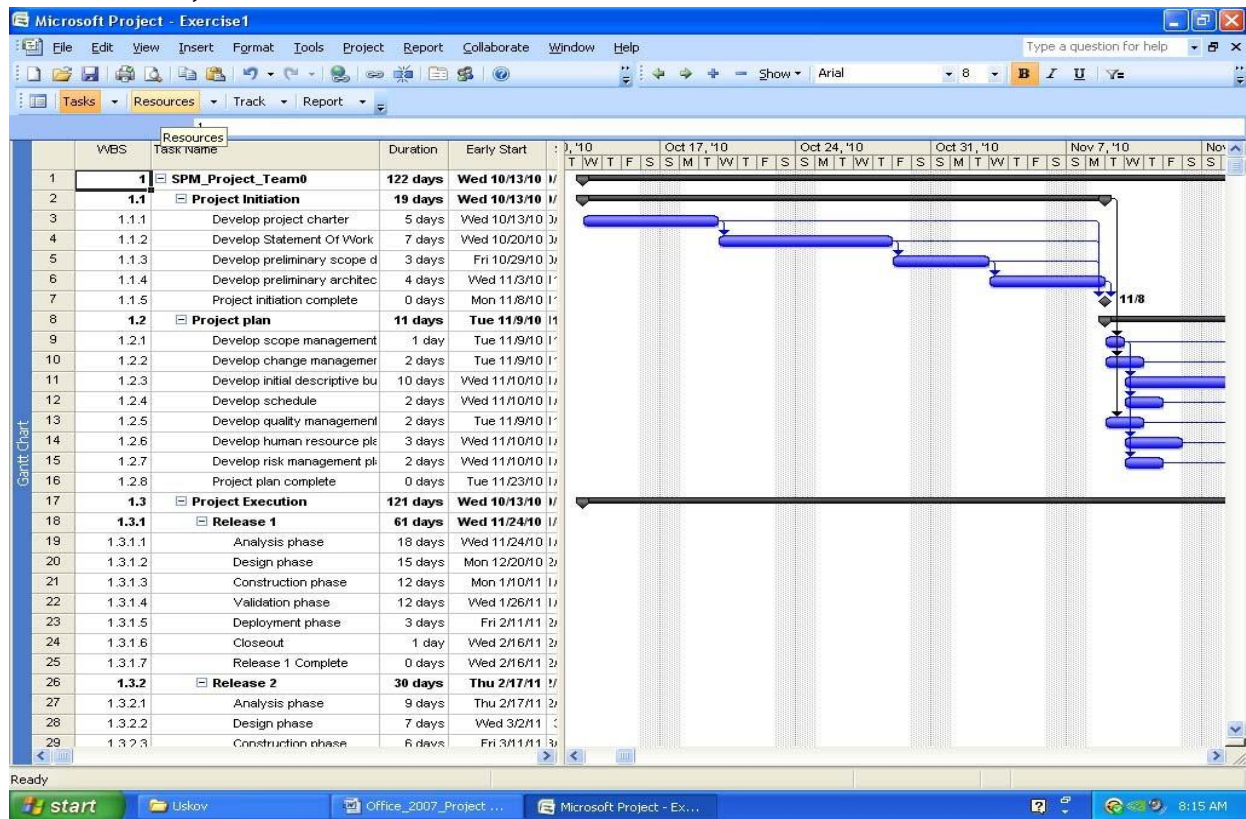
Navigate to View->Resource Sheet.



The screenshot shows the 'Resource Sheet' view in Microsoft Project. The table has the following columns: Resource Name, Type, Material Label, Initials, Group, Max. Units, Std. Rate, Ovt. Rate, Cost/Use, Accrue At, Base Calendar, and Code. The table is currently empty.

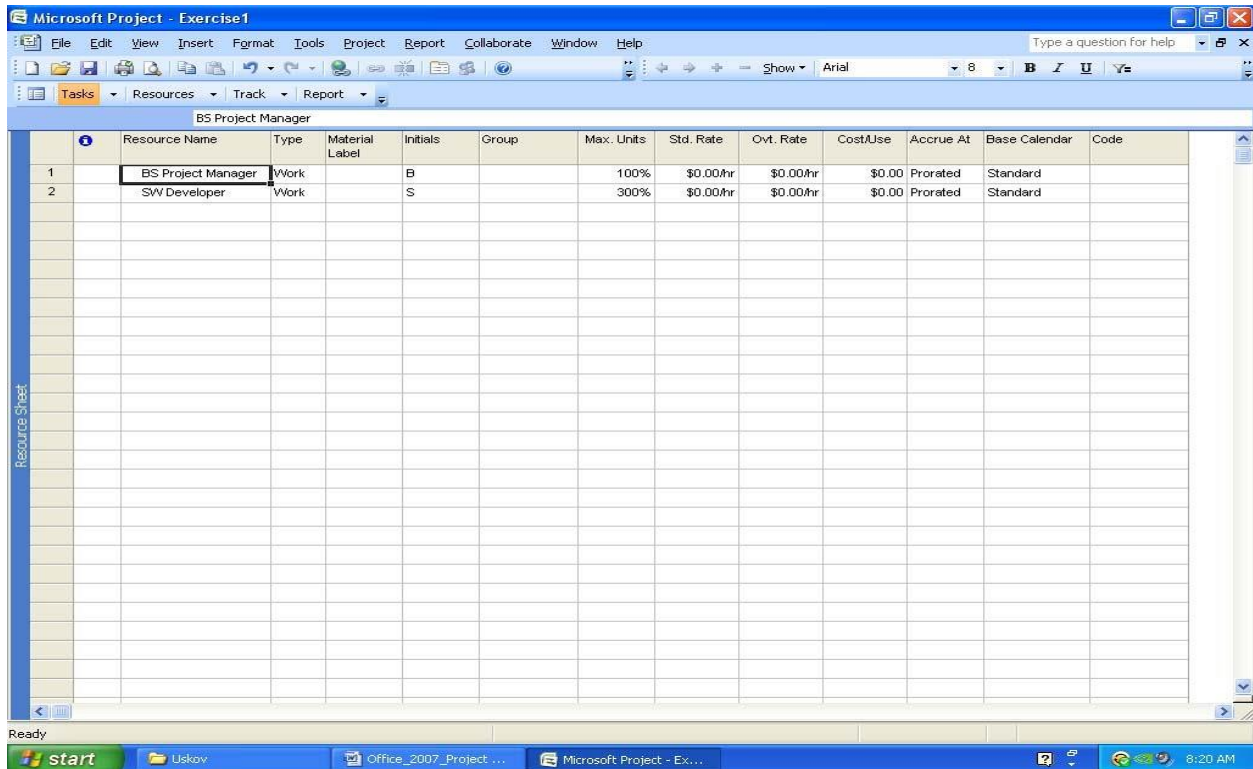
Resource Name	Type	Material Label	Initials	Group	Max. Units	Std. Rate	Ovt. Rate	Cost/Use	Accrue At	Base Calendar	Code
---------------	------	----------------	----------	-------	------------	-----------	-----------	----------	-----------	---------------	------

Or click on Project Guide Toolbar->Resources

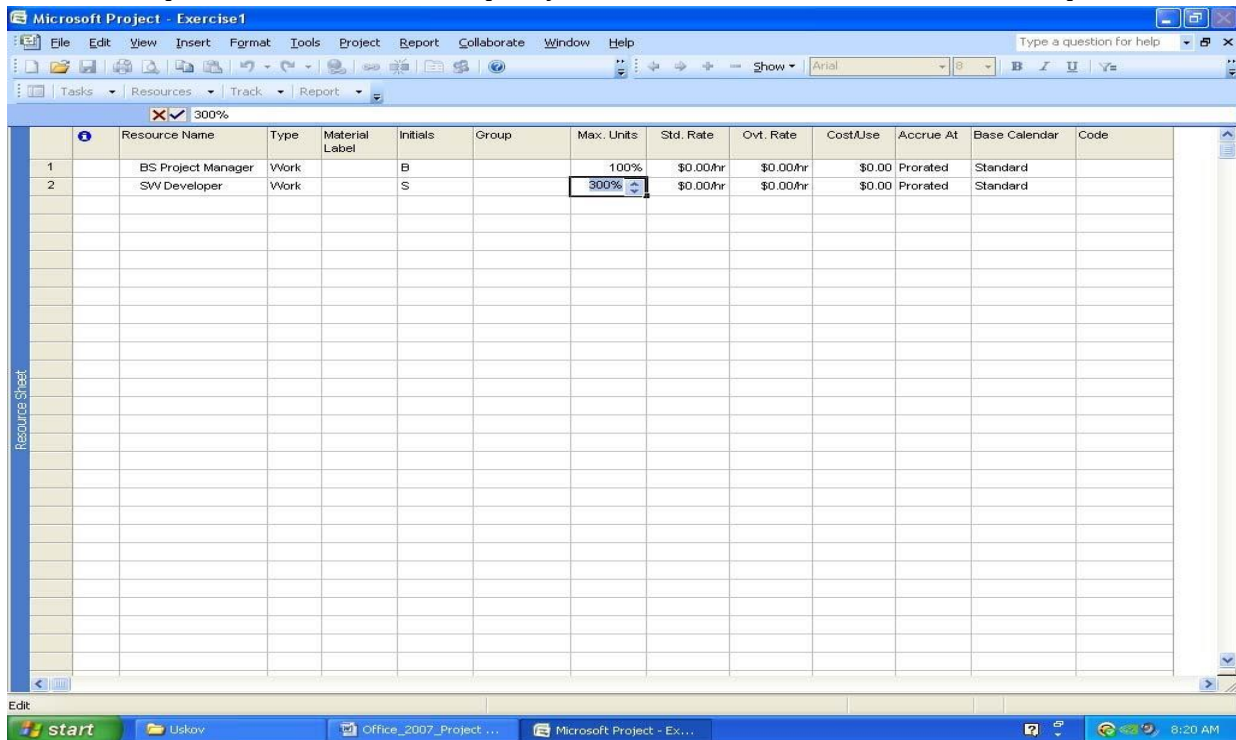


2.2 ADDING RESOURCE ATTRIBUTES

Resource Name is the title of resource to be used

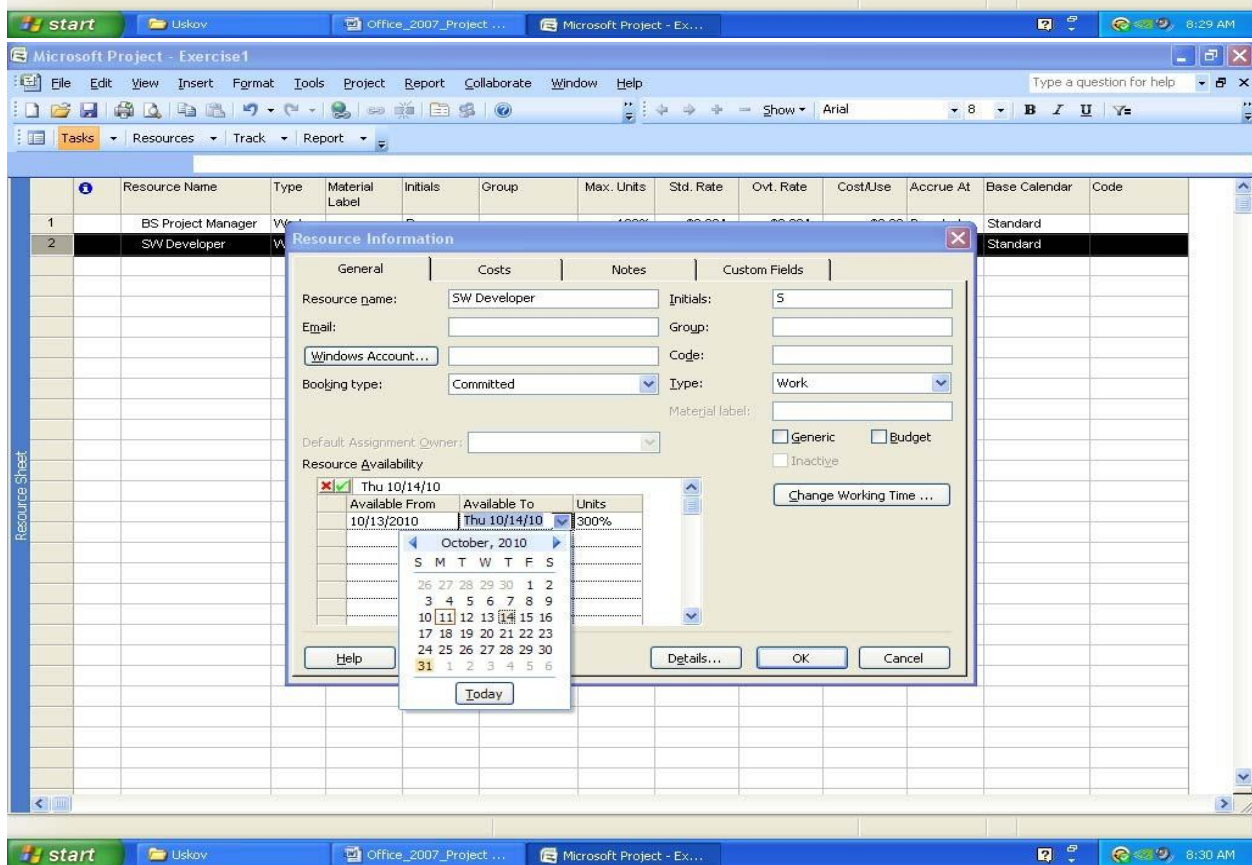
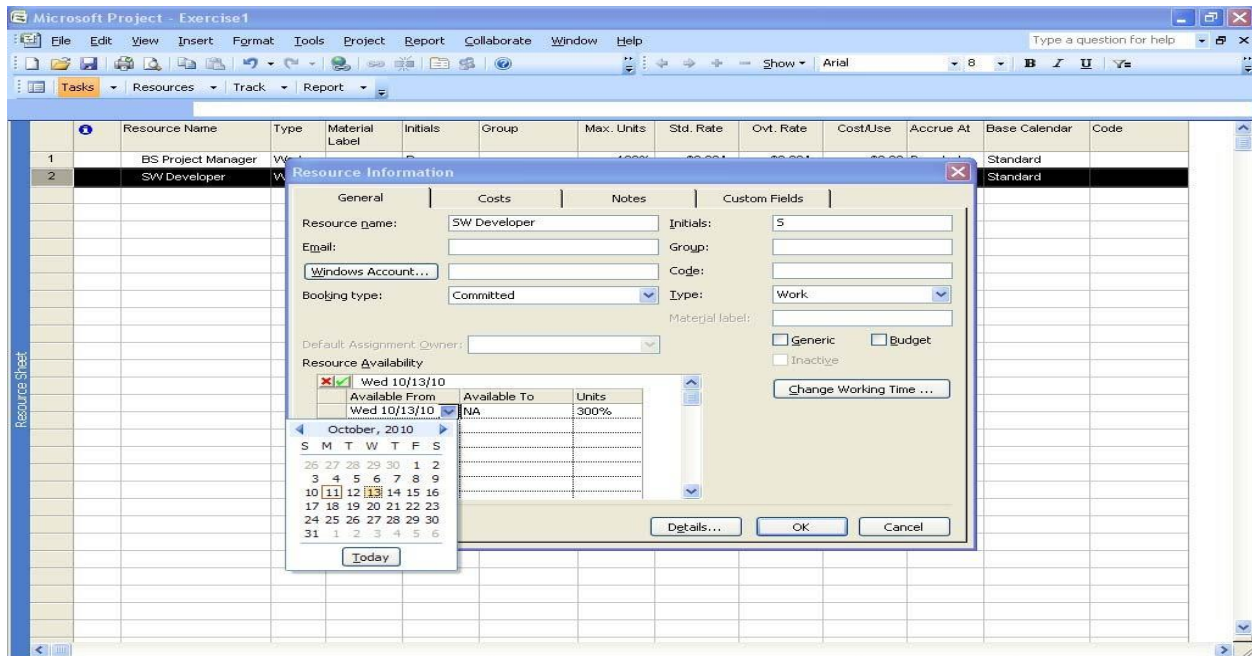


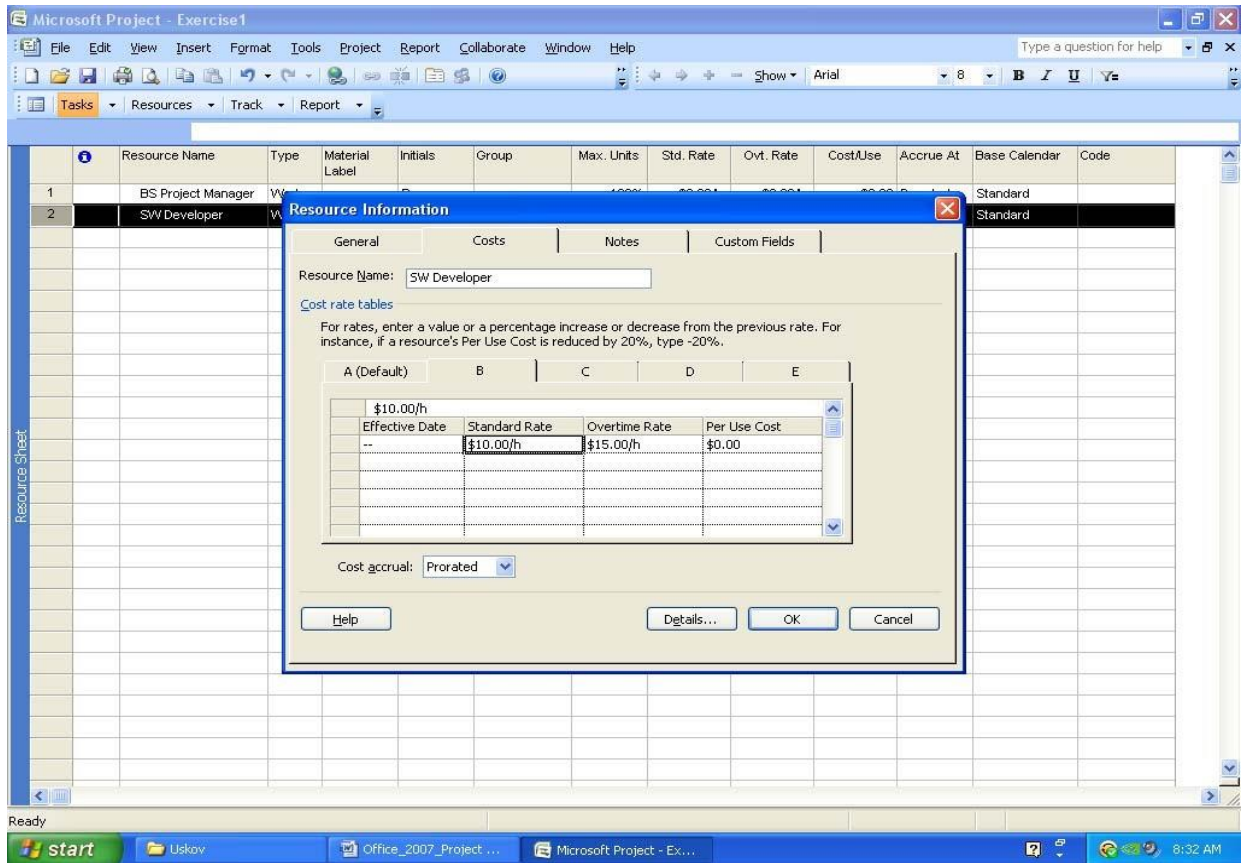
Max Units represents the maximum capacity for which resource is available to accomplish the task.



2.3 Entering Multiple Pay Rates

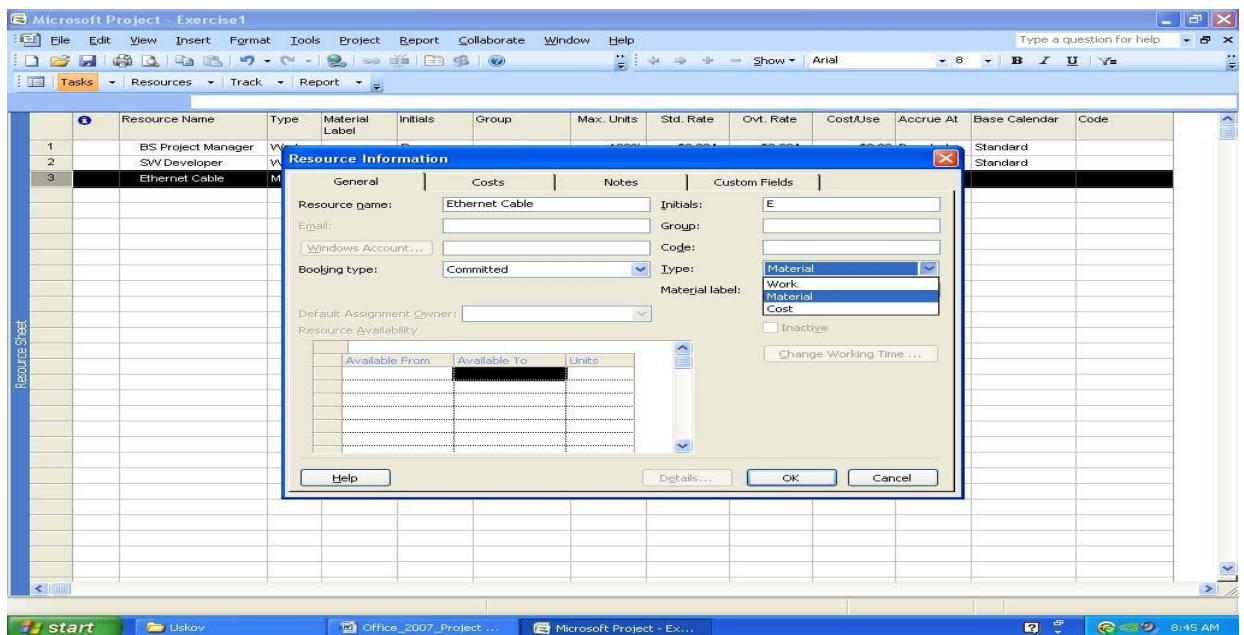
Double-clicking the resource and displaying the Resource Information Screen->Costs tab. Changing the rates for a given date range or specifying another pay rate for another calendar

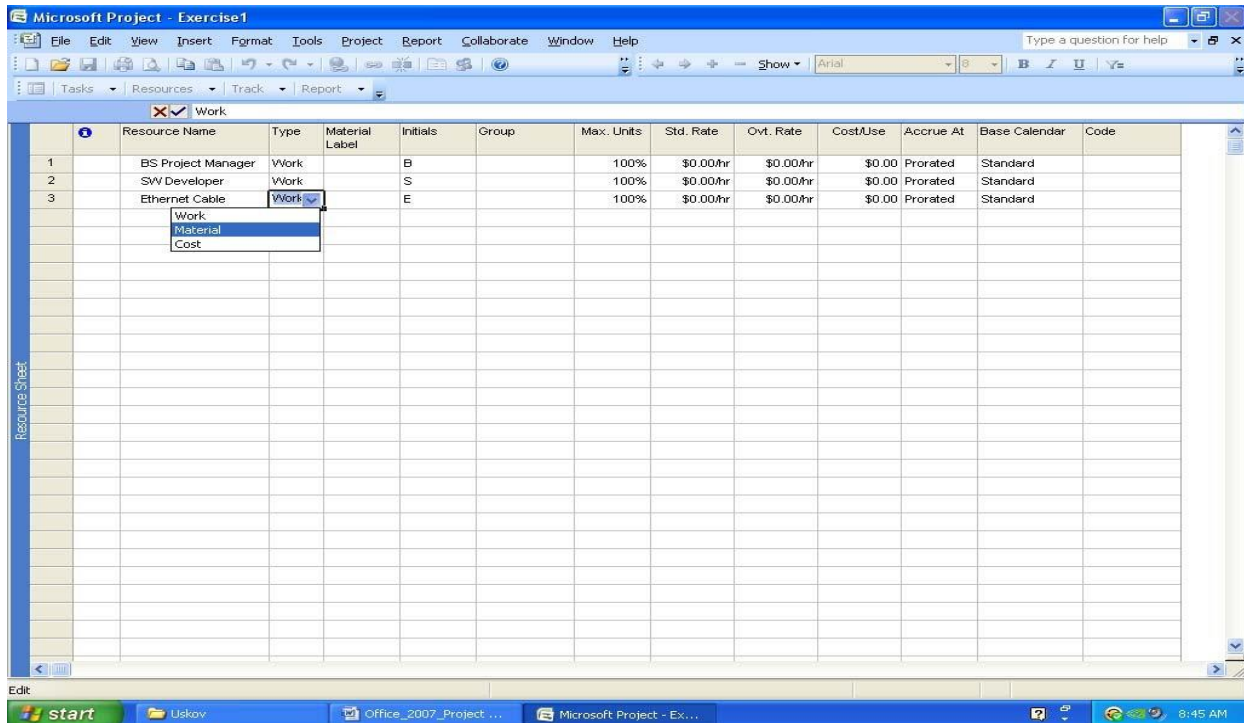




2.4 Entering Equipment and Material Resources

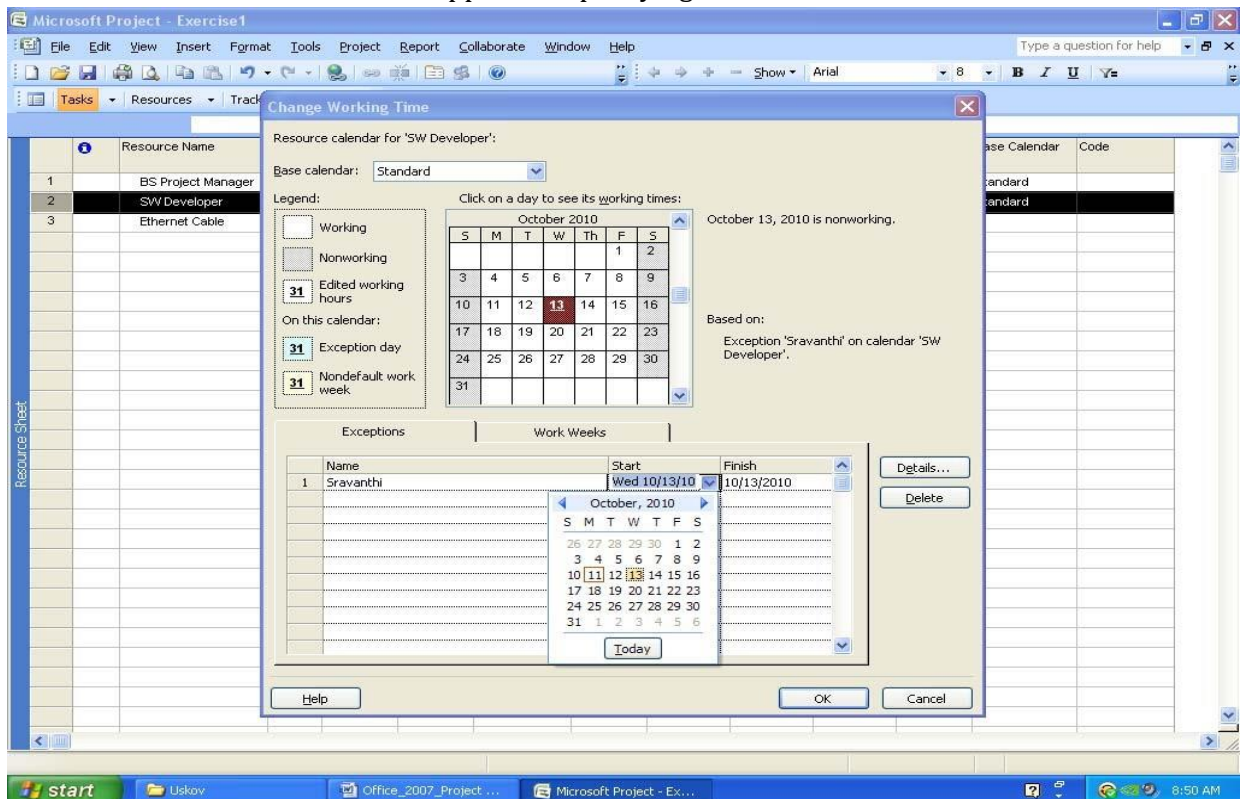
Double-Click an open row on resource sheet. Click a drop-down box in the Type Field and select Material

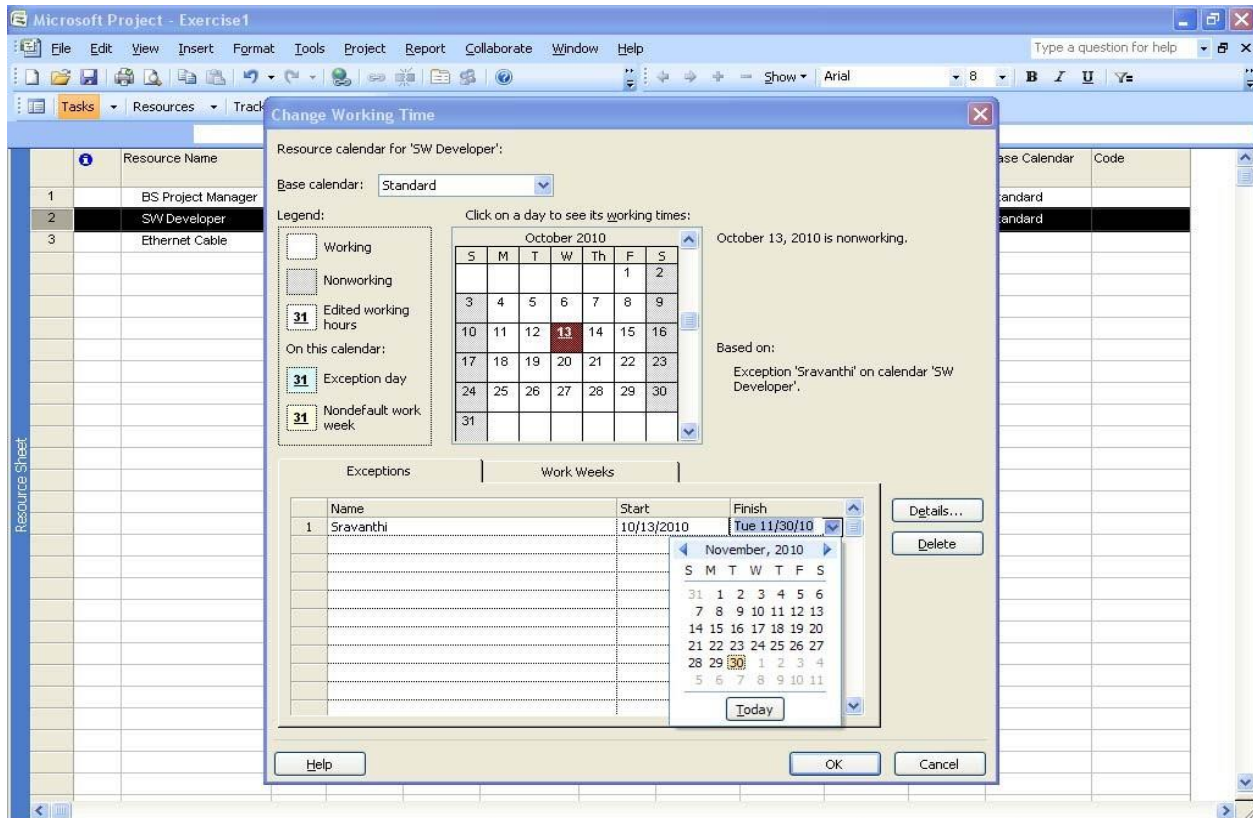




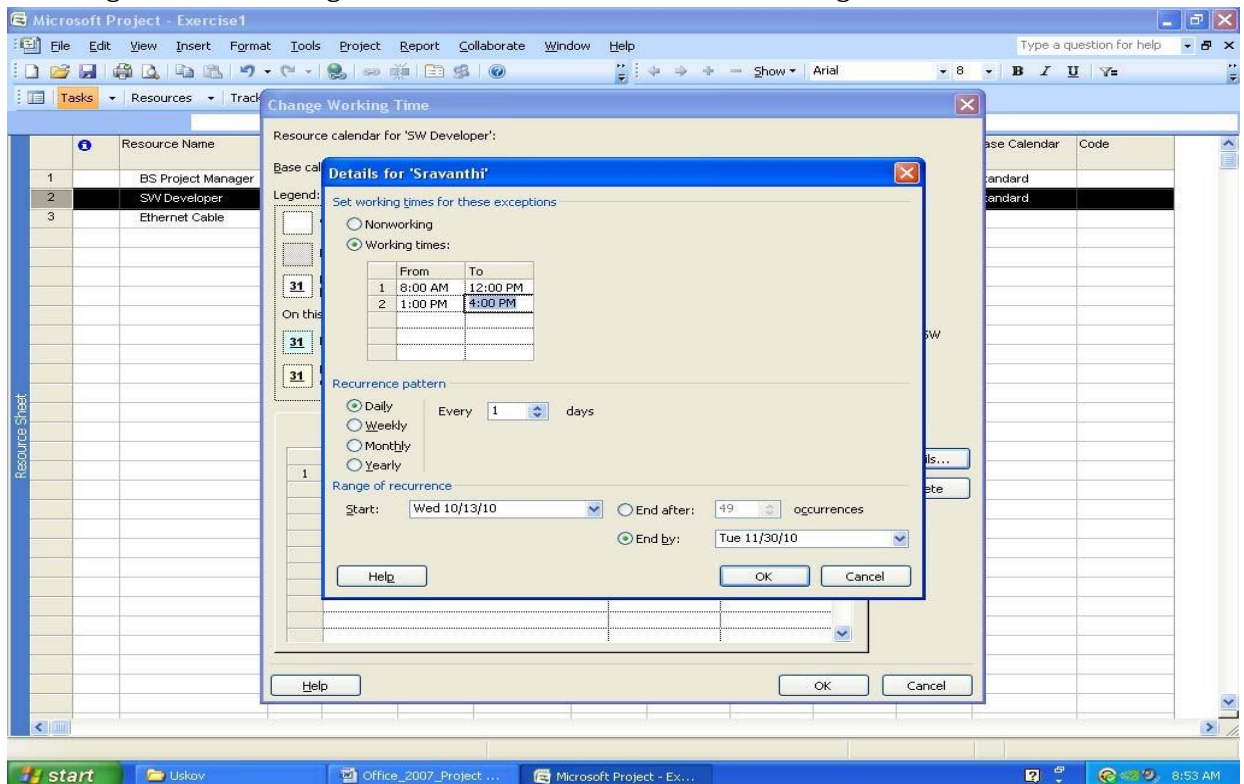
2.5 Editing the Resource Calendar

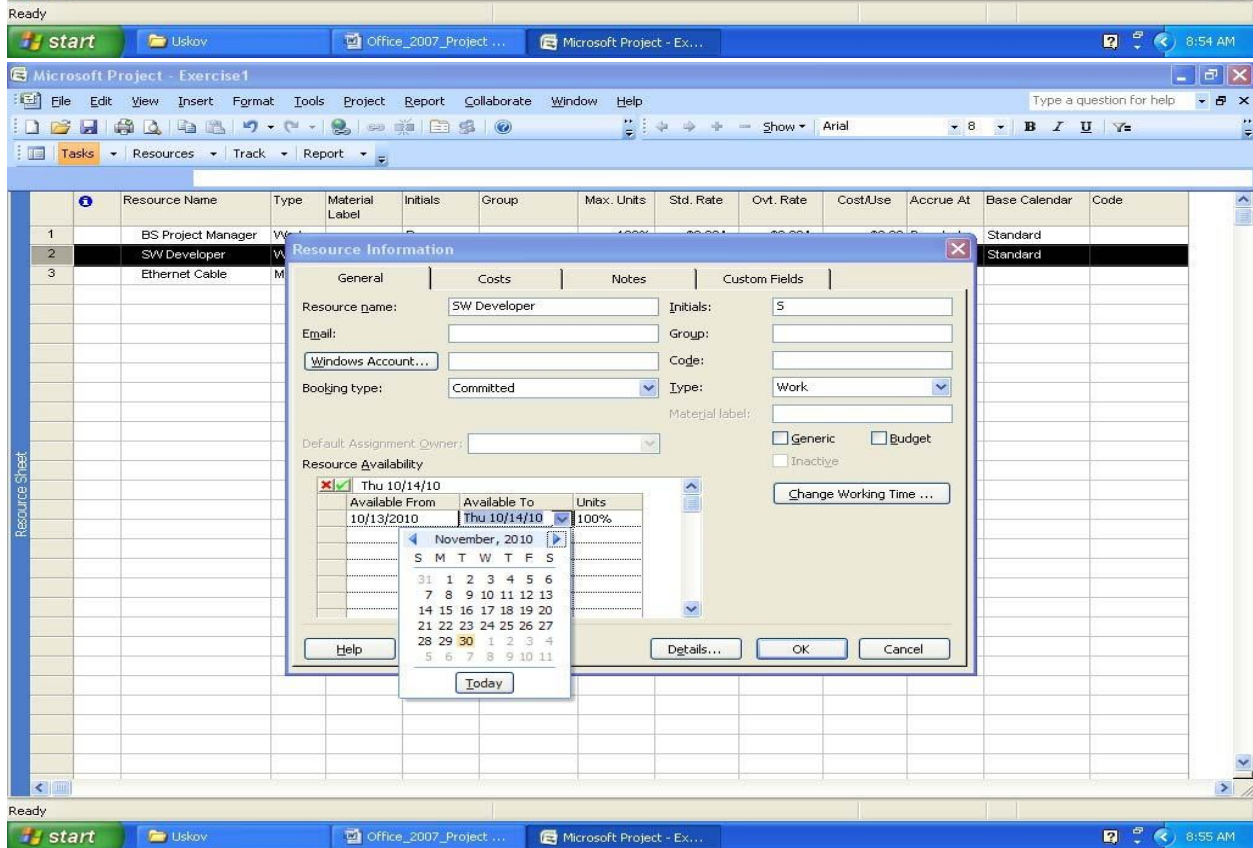
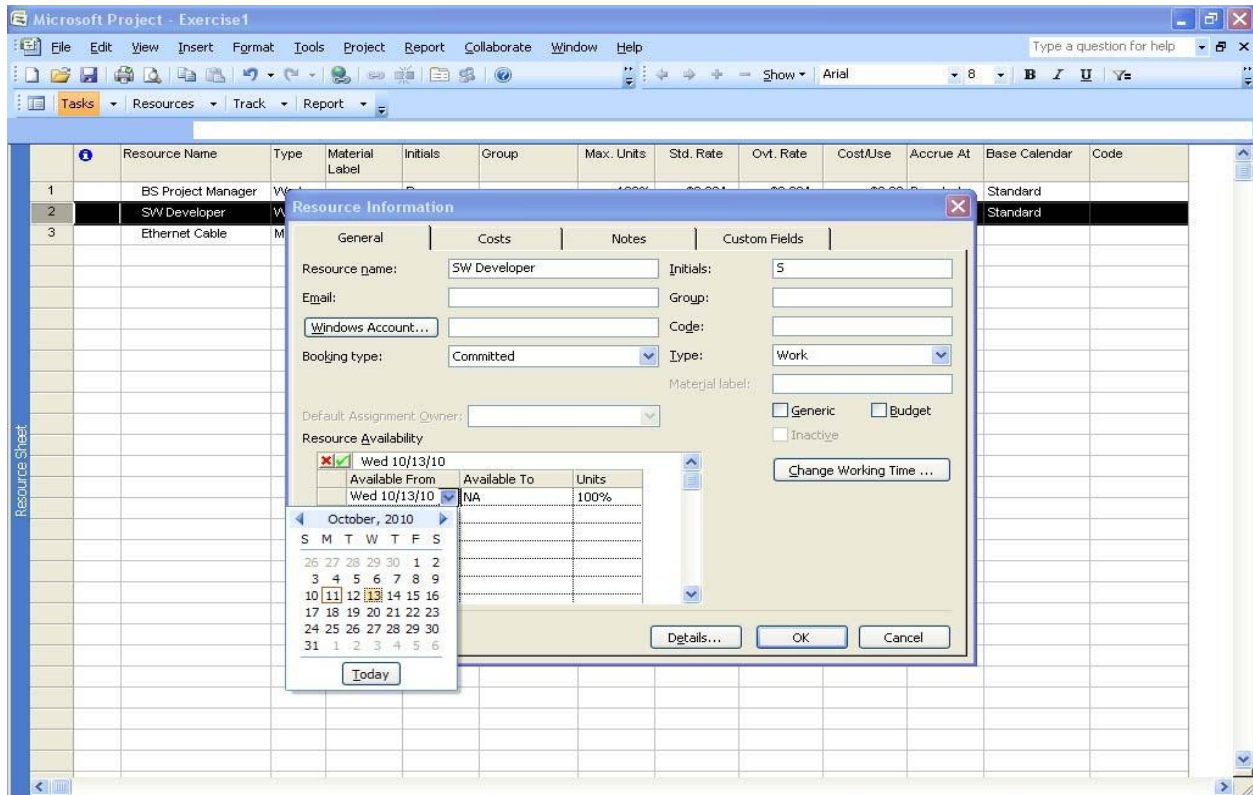
You can add specific working times and days by double-clicking the resource, selecting the Working Time Tab in the new window that appears & specifying the new work times.





To change the work timings click on details button-> select working times as shown below



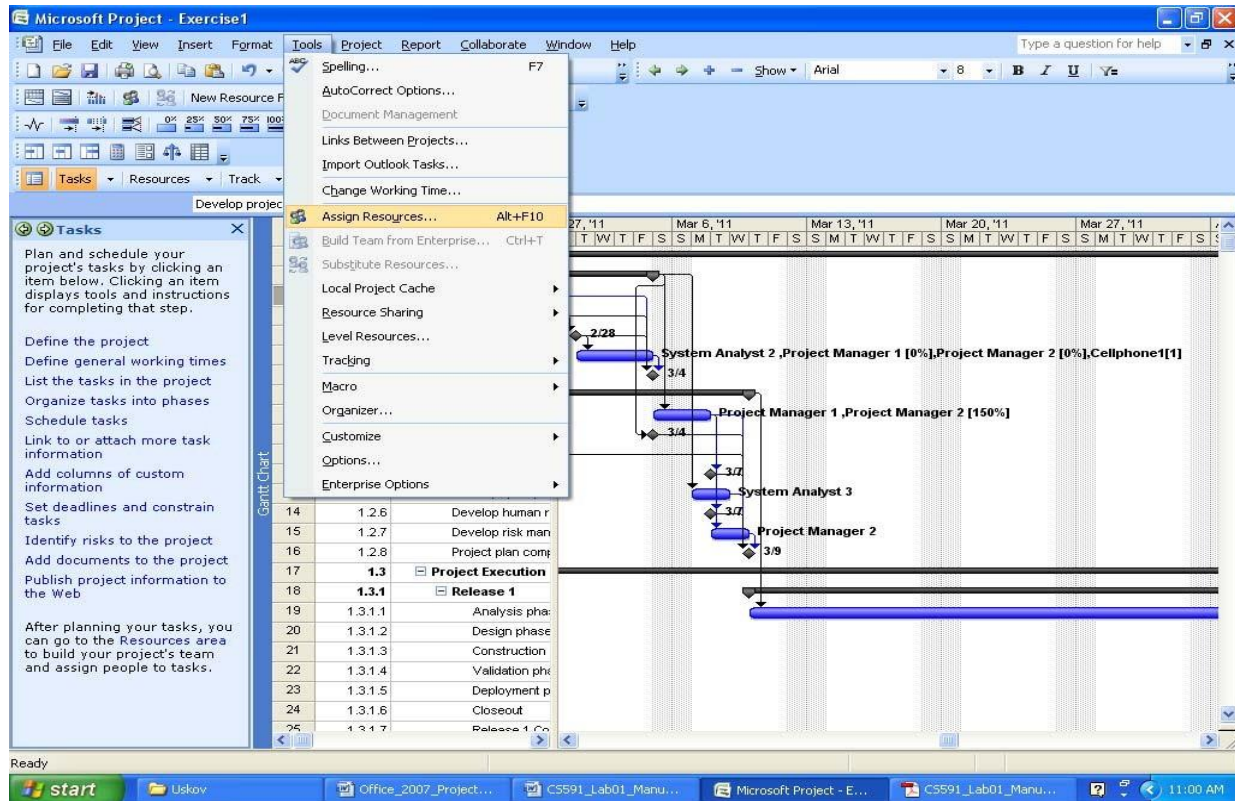


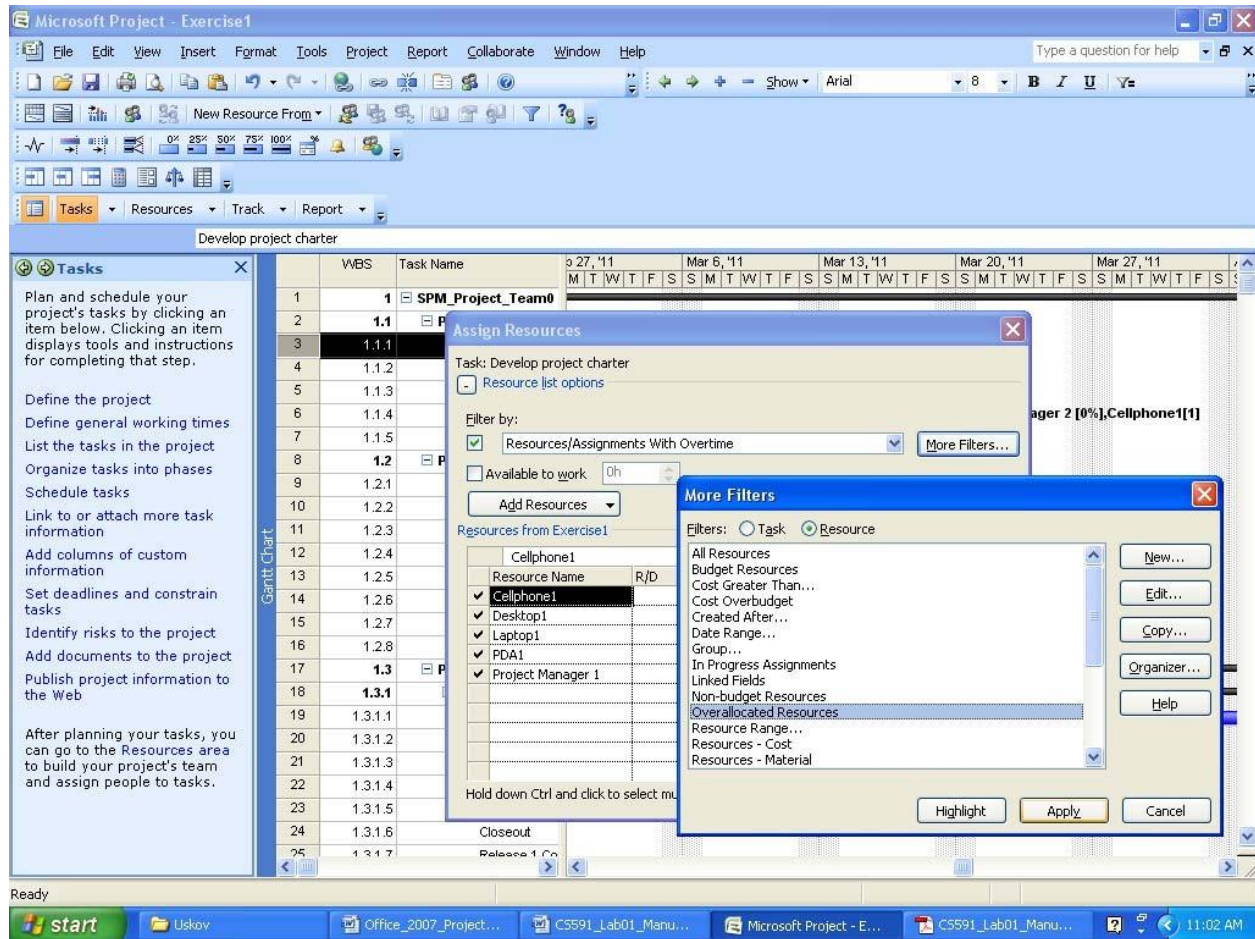
2.6 Dealing With Over Allocated Resources

Navigate to View ->Resource Usage to determine whether a resource is over allocated. Now, navigate to task->Double-click on task name -> Resources tab to assign resources to the task.

The screenshot displays the Microsoft Project - Exercise1 interface. The main window shows a Gantt Chart with tasks listed in a table. The Task Information dialog box is open, showing the Resources tab. The task 'Develop project charter' is selected, with a duration of 5 days. The Resources tab shows a list of resources assigned to the task, including 'BS Project Manager' (100% units, \$0.00 cost) and 'Ethernet Cable' (100% units, \$0.00 cost). The dialog box also includes tabs for General, Predecessors, Resources, Advanced, Notes, and Custom Fields. The background Gantt Chart shows a project timeline from October 17, 2010, to November 7, 2010, with various tasks and their durations.

WBS	Task Name	Duration	Early Start
1	1 SPM_Project_Team0	120 days	Wed 10/13/10
2	1.1 Project Initiation	17 days	Wed 10/13/10
3	1.1.1 Develop		
4	1.1.2 Develop		
5	1.1.3 Develop		
6	1.1.4 Develop		
7	1.1.5 Project		
8	1.2 Project		
9	1.2.1 Develop		
10	1.2.2 Develop		
11	1.2.3 Develop		
12	1.2.4 Develop		
13	1.2.5 Develop		
14	1.2.6 Develop		
15	1.2.7 Develop		
16	1.2.8 Project		
17	1.3 Project		
18	1.3.1 Release		
19	1.3.1.1 Analysis		
20	1.3.1.2 Design		
21	1.3.1.3 Construction		
22	1.3.1.4 Deployment		
23	1.3.1.5 Closeout		
24	1.3.1.6 Release 1 Complete		
25	1.3.2 Release 2	30 days	Tue 2/15/11
26	1.3.2.1 Analysis phase	9 days	Tue 2/15/11
27	1.3.2.2 Design phase	7 days	Mon 2/28/11
28	1.3.2.3 Construction phase	6 days	Wed 3/9/11





2.7 LEVELING RESOURCES

Navigate to Task->tools->Level Resources. Please check the Exercise for the leveling Resources.

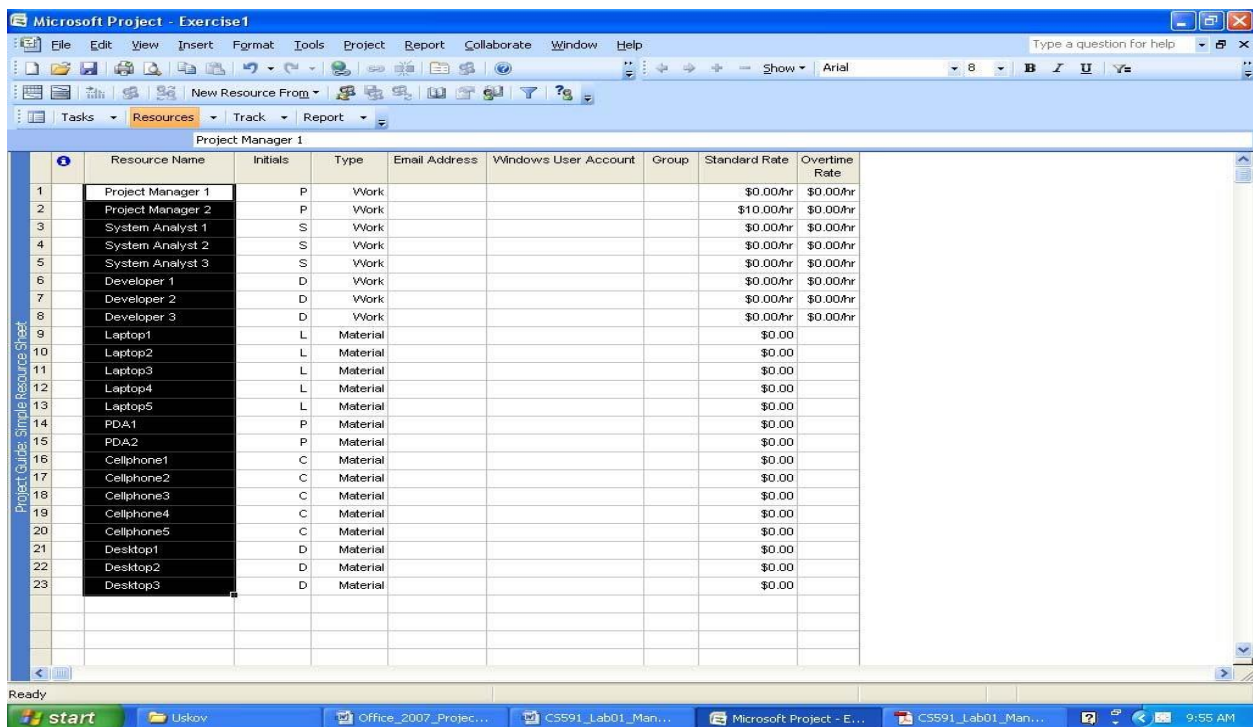
2.8 Exercises

2.8.1 Assigning Human and Material resources to tasks

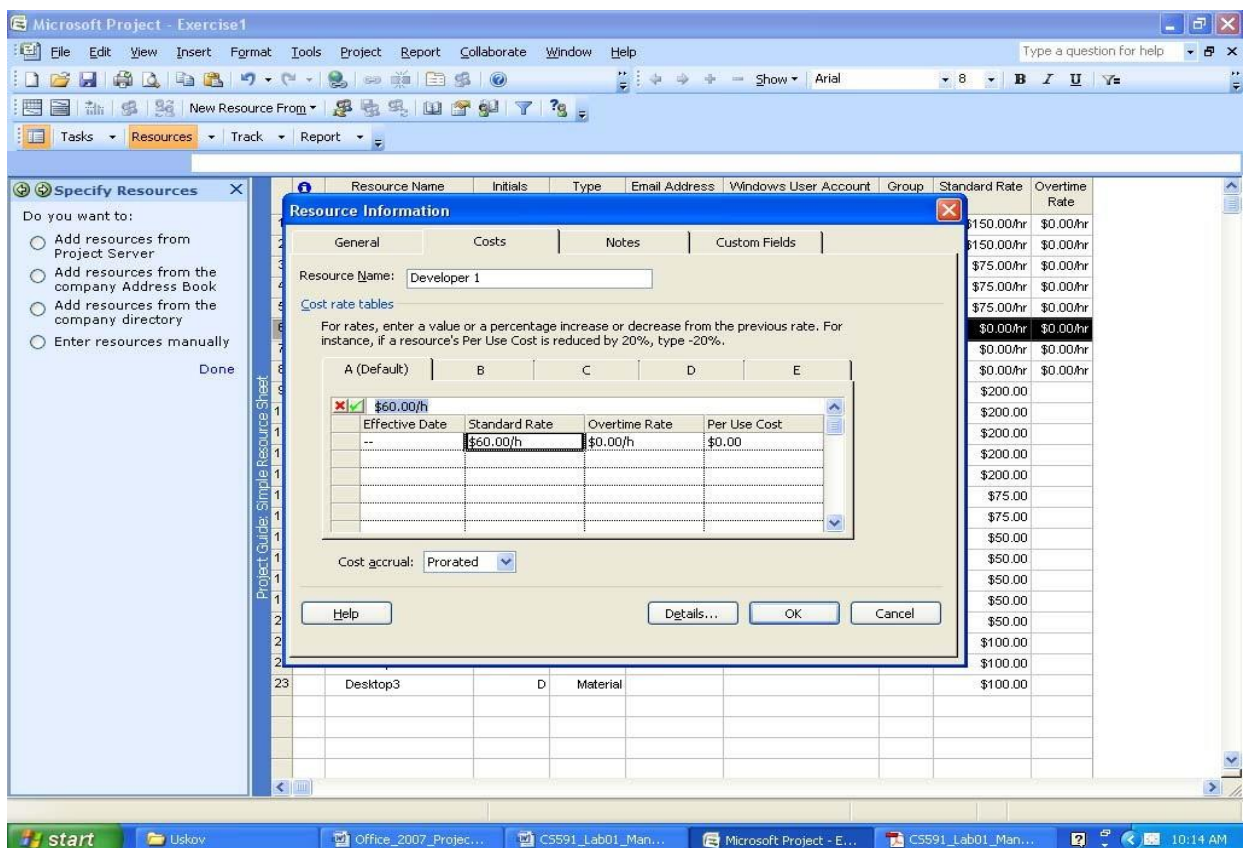
	Task No	Human Resources	Material Resources
Column 0	Column 1	Column 5	Column 6
1	1		
2	1.1		
3	1.1.1	Project Manager 1	Laptop1, PDA1,Cellphone1
4	1.1.2	Project Manager 2	Laptop2, PDA2,Cellphone2
5	1.1.3	Project Manager 2, System Analyst 1	Laptop2, PDA2,Cellphone2, Laptop3, Cellphone3
6	1.1.4	System Analyst 1, System Analyst 2	Laptop3, Cellphone 3, Laptop4, Cellphone 4
7	1.1.5	Project Manager 1, Project	Laptop1, PDA1,Cellphone1,

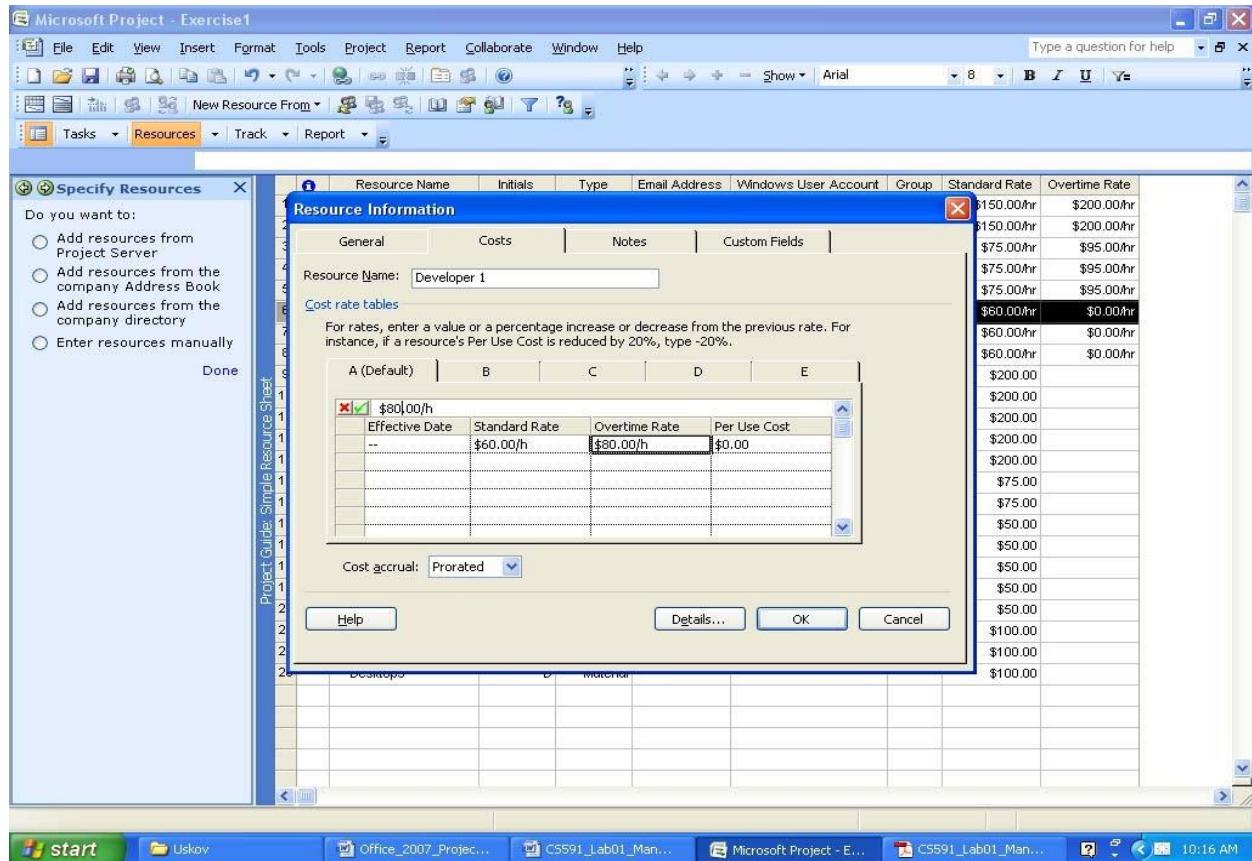
		Manager 2	Laptop2, PDA2,Cellphone 2
8	1.2		
9	1.2.1	Project Manager 1	Laptop1, PDA1,Cellphone1
10	1.2.2	Project Manager 1, System Analyst 1	Laptop1, PDA1,Cellphone1, Laptop3, Cellphone3
11	1.2.3	System Analyst 1	Laptop3, Cellphone3
12	1.2.4	System Analyst 2	Laptop4, Cellphone4
13	1.2.5	System Analyst 1, System Analyst 2	Laptop3, Cellphone3, Laptop4, Cellphone4
14	1.2.6	System Analyst 3	Laptop5, Cellphone5
15	1.2.7	System Analyst 2, System Analyst 3	Laptop4, Cellphone4, Laptop5, Cellphone5
16	1.2.8	Project Manager 1	Laptop1, PDA1,Cellphone1
17	1.3		
18	1.3.1		
19	1.3.1.1	Project Manager 1	Laptop1,PDA1,Cellphone1
20	1.3.1.2	Developer 1, Developer 2	Desktop1,Desktop2
21	1.3.1.3	Developer 2, Developer 3	Desktop2,desktop3
22	1.3.1.4	Developer 3	Desktop3
23	1.3.1.5	Developer1, Developer 3	Desktop1,Desktop3
24	1.3.1.6	System Analyst 1	Laptop3,Cellphone3
25	1.3.1.7	Developer 1, Project Manager 1	Laptop1,PDA1,cellphone1,Desktop1
26	1.3.2		
27	1.3.2.1	Project Manager 1	Laptop1,PDA1,Cellphone1
28	1.3.2.2	Developer 1, Developer 2	Desktop1,Desktop2
29	1.3.2.3	Developer 2, Developer 3	Desktop2,desktop3
30	1.3.2.4	Developer 3	Desktop3
31	1.3.2.5	Developer1, Developer 3	Desktop1,Desktop3
32	1.3.2.6	System Analyst 1	Laptop3,Cellphone3
33	1.3.2.7	Developer 1, Project Manager 1	Laptop1,PDA1,cellphone1,Desktop1
34	1.3.3		
35	1.4	Project Manager 2	Laptop2,PDA2,Cellphone2
36	1.5	Project Manager 1	Laptop1,PDA1,Cellphone1

2.8.1.1 Enter the list of resources

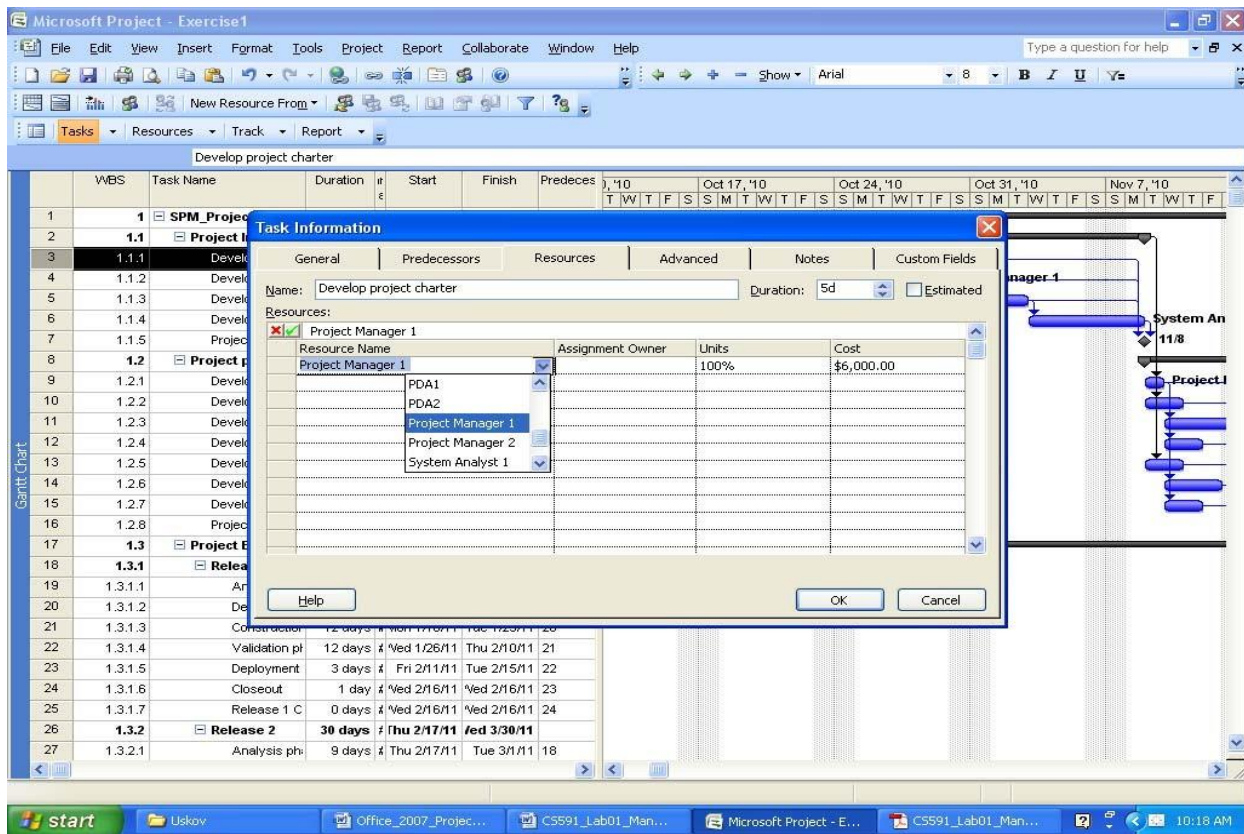


2.8.1.2 Entering Cost for each Resource

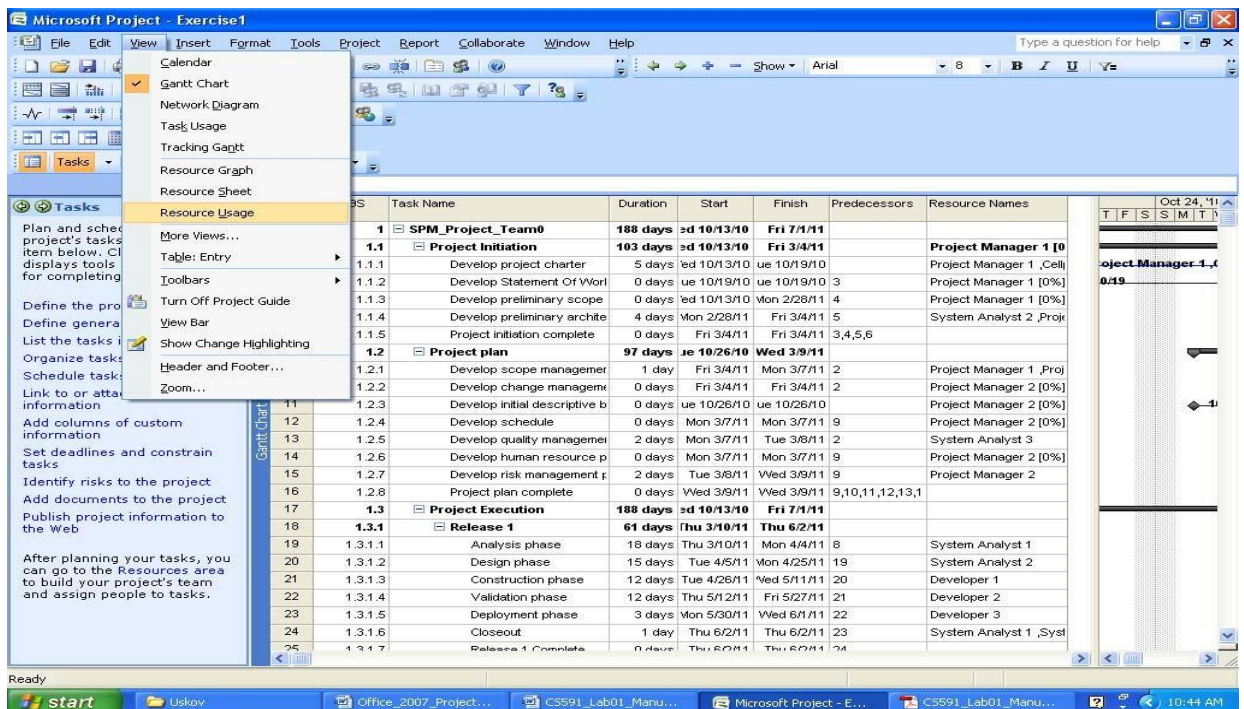




2.8.1.3 Assigning each resource to the task



2.8.1.4 Resource Usage



2.8.1.5 Dealing with over allocated Resource

Navigate to task->Assign Resources

The screenshot shows the Microsoft Project - Exercise1 window with the Resources sheet selected. The Resources sheet lists various resources and their availability. The Resources sheet is divided into two sections: 'Specify Resources' and 'Resources'.

Specify Resources

- Do you want to:
 - ☐ Add resources from Project Server
 - ☐ Add resources from the company Address Book
 - ☐ Add resources from the company directory
 - ☐ Enter resources manually

Resources

Resource Name	Start	Initials	Type	Email Address	Windows User Account	Group	Standard Rate	Overtime Rate
Project Manager 1	Wed 10/13/10	P	Work				\$150.00/hr	\$200.00/hr
Project Manager 2	Wed 10/13/10	P	Work				\$150.00/hr	\$200.00/hr
System Analyst 1	Thu 3/10/11	S	Work				\$75.00/hr	\$95.00/hr
System Analyst 2	Tue 3/1/11	S	Work				\$75.00/hr	\$95.00/hr
System Analyst 3	Mon 3/7/11	S	Work				\$75.00/hr	\$95.00/hr
Developer 1	Tue 4/26/11	D	Work				\$60.00/hr	\$80.00/hr
Developer 2	Thu 5/12/11	D	Work				\$60.00/hr	\$80.00/hr
Developer 3	Mon 5/30/11	D	Work				\$60.00/hr	\$80.00/hr
Laptop1	Wed 10/13/10	L	Material				\$200.00	
Laptop2	NA	L	Material				\$200.00	
Laptop3	NA	L	Material				\$200.00	
Laptop4	Wed 6/8/11	L	Material				\$200.00	
Laptop5	NA	L	Material				\$200.00	
PDA1	Wed 10/13/10	P	Material				\$75.00	
PDA2	NA	P	Material				\$75.00	
Cellphone1	Tue 3/1/11	C	Material				\$50.00	
Cellphone2	NA	C	Material				\$50.00	
Cellphone3	NA	C	Material				\$50.00	
Cellphone4	NA	C	Material				\$50.00	
Cellphone5	NA	C	Material				\$50.00	
Desktop1	NA	D	Material				\$100.00	
Desktop2	NA	D	Material				\$100.00	
Desktop3	NA	D	Material				\$100.00	

The screenshot shows the Microsoft Project - Exercise1 window with the Gantt Chart view selected. The Tools menu is open, showing the 'Assign Resources...' option. The Gantt Chart shows a project schedule with tasks and resources assigned.

Tools

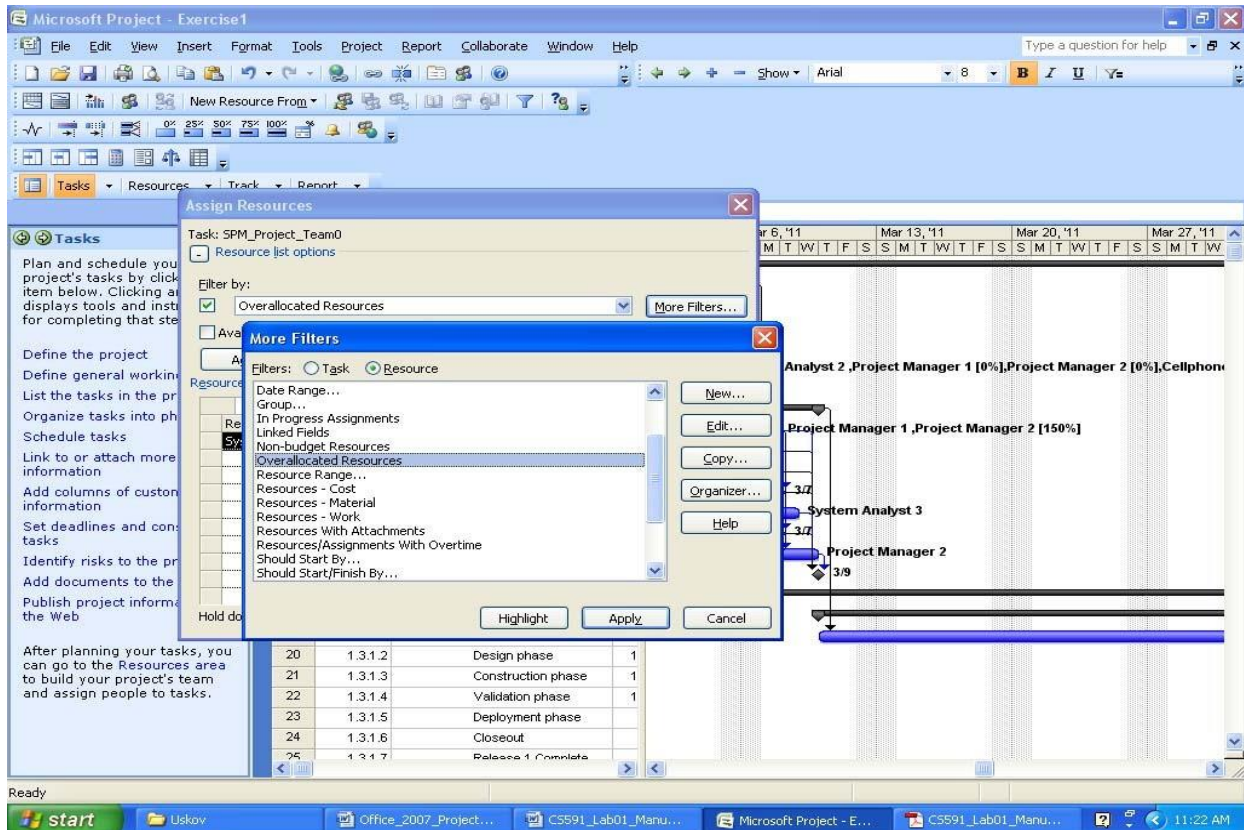
- Spelling...
- AutoCorrect Options...
- Document Management
- Links Between Projects...
- Import Outlook Tasks...
- Change Working Time...
- Assign Resources...** (Alt+F10)
- Build Team from Enterprise...
- Substitute Resources...
- Local Project Cache
- Resource Sharing
- Level Resources...
- Tracking
- Macro
- Organizer...
- Customize
- Options...
- Enterprise Options

Gantt Chart

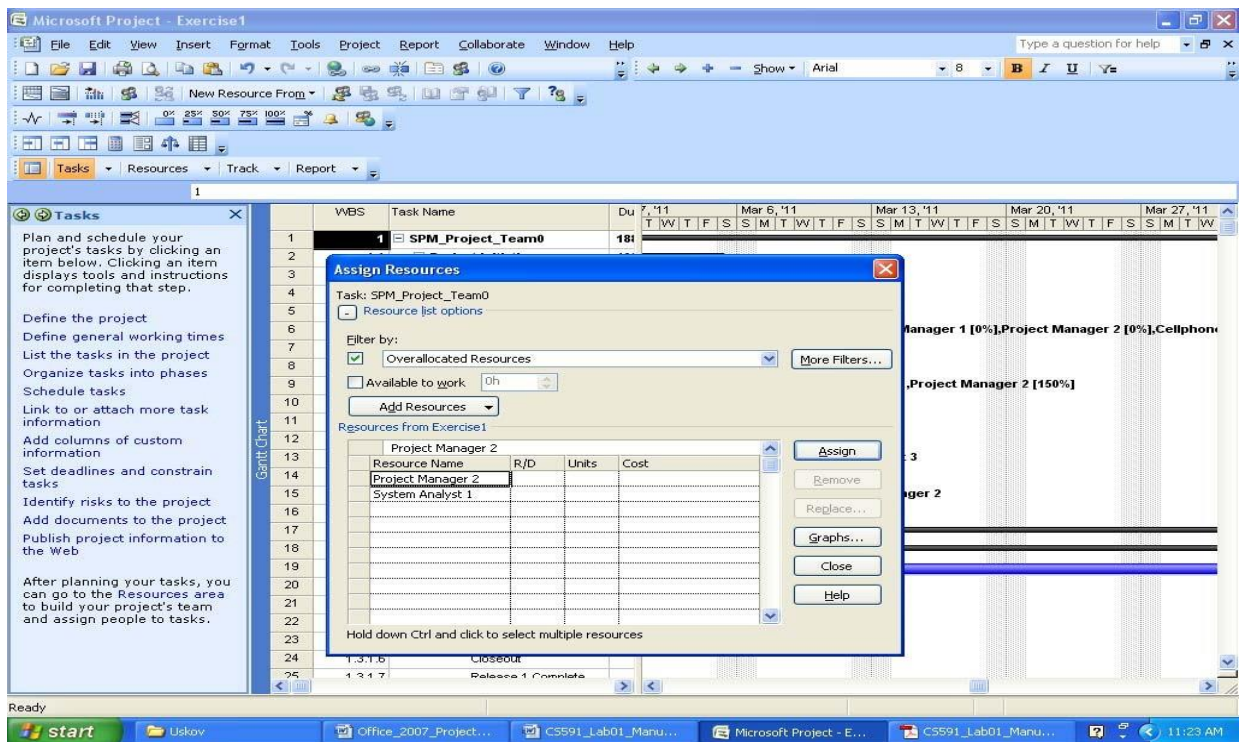
The Gantt Chart shows a project schedule with tasks and resources assigned. The tasks are listed in the left pane, and the Gantt Chart shows the duration of each task and the resources assigned to it. The resources are listed in the right pane, and the Gantt Chart shows the duration of each resource and the tasks assigned to it.

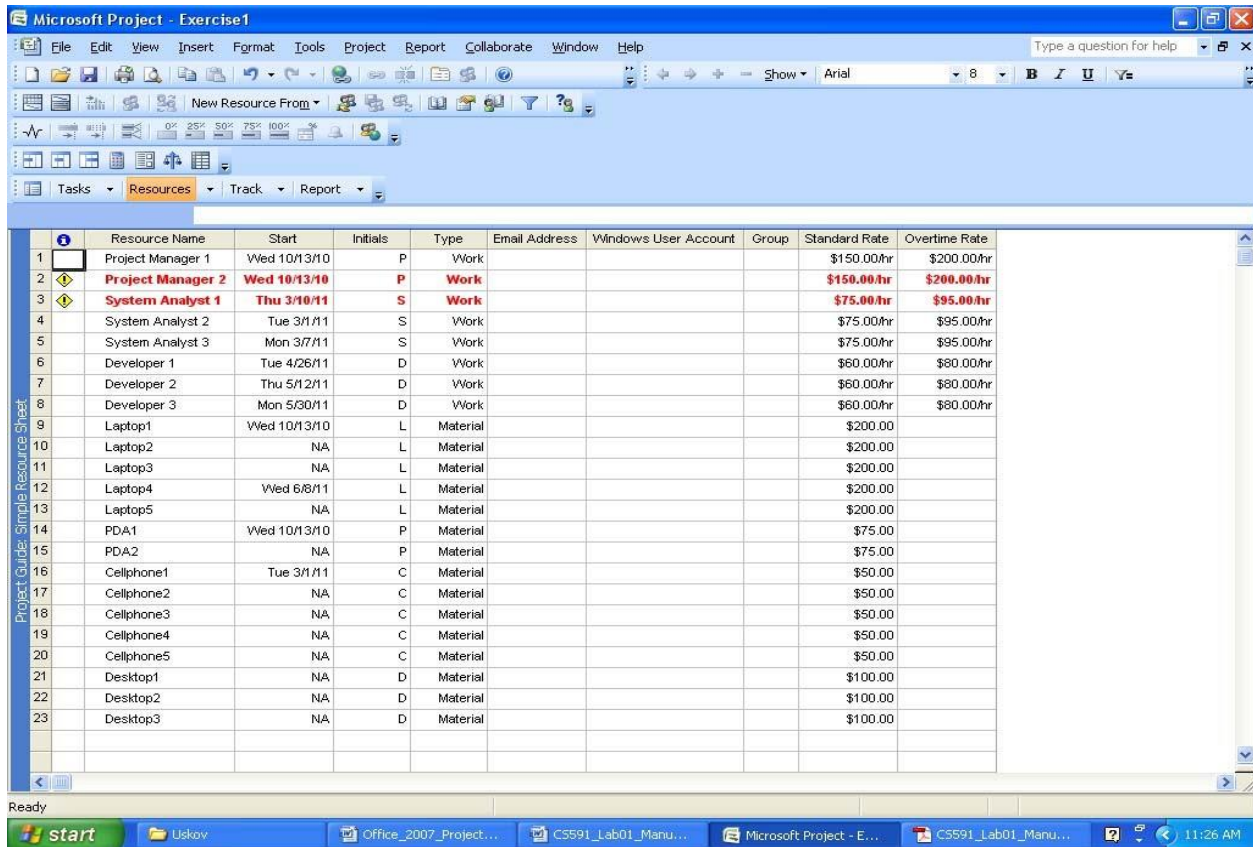
Task List

Task ID	Task Name	Duration
14	1.2.6 Develop human resource p	1
15	1.2.7 Develop risk management p	1
16	1.2.8 Project plan complete	1
17	1.3 Project Execution	18
18	1.3.1 Release 1	6
19	1.3.1.1 Analysis phase	1
20	1.3.1.2 Design phase	1
21	1.3.1.3 Construction phase	1
22	1.3.1.4 Validation phase	1
23	1.3.1.5 Deployment phase	1
24	1.3.1.6 Closeout	1
25	1.3.1.7 Release 1 Complete	1

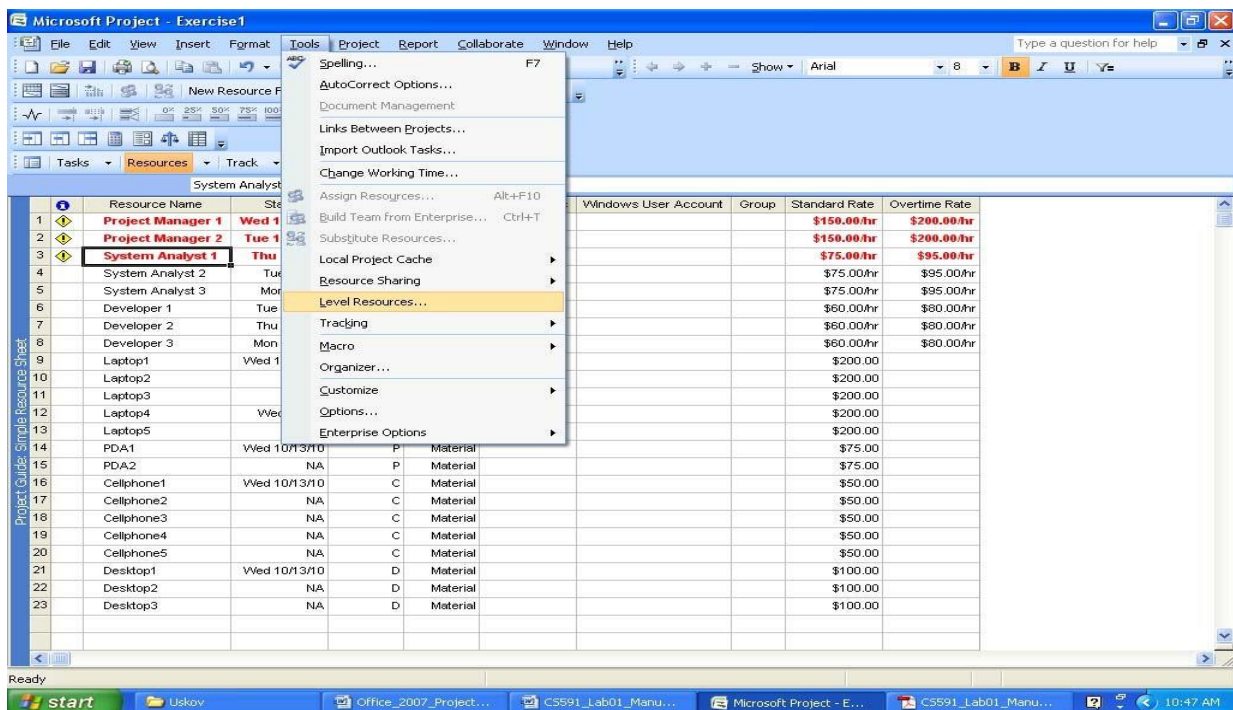


After clicking apply button. Again Navigate from tools->Assign Resources to display which Resources are over allocated





2.8.1.6 Leveling Resources



Microsoft Project - Exercise1

File Edit View Insert Format Tools Project Report Collaborate Window Help

Project Guide: Simple Resource Sheet

Resource Name	Start	Overtime Rate
Project Manager 1	Wed 10/13/10	\$200.00/hr
Project Manager 2	Tue 10/26/10	\$200.00/hr
System Analyst 1	Thu 3/10/11	\$95.00/hr
System Analyst 2	Tue 3/1/11	\$95.00/hr
System Analyst 3	Mon 3/7/11	\$95.00/hr
Developer 1	Tue 4/26/11	\$80.00/hr
Developer 2	Thu 5/12/11	\$80.00/hr
Developer 3	Mon 5/30/11	\$80.00/hr
Laptop1	Wed 10/13/10	\$80.00/hr
Laptop2	NA	
Laptop3	NA	
Laptop4	Wed 6/8/11	
Laptop5	NA	
PDA1	Wed 10/13/10	
PDA2	NA	
Cellphone1	Wed 10/13/10	
Cellphone2	NA	
Cellphone3	NA	
Cellphone4	NA	
Cellphone5	NA	
Desktop1	Wed 10/13/10	
Desktop2	NA	
Desktop3	NA	

Resource Leveling

Leveling calculations

☐ Automatic ☒ Manual

Look for overallocations on a Day by Day basis

☒ Clear leveling values before leveling

Leveling range for 'Exercise1'

☒ Level entire project

☐ Level From: Wed 10/13/10 To: Fri 7/1/11

Resolving overallocations

Leveling order: Standard

☐ Level only within available slack

☒ Leveling can adjust individual assignments on a task

☒ Leveling can create splits in remaining work

☐ Level resources with the proposed booking type

Help Clear Leveling... Level Now OK Cancel

start Uskov Office_2007_Project... CS591_Lab01_Manu... Microsoft Project - E... CS591_Lab01_Manu... 10:47 AM

Microsoft Project - Exercise1

File Edit View Insert Format Tools Project Report Collaborate Window Help

Project Guide: Simple Resource Sheet

Resource Name	Start	Initials	Type	Email Address	Windows User Account	Group	Standard Rate	Overtime Rate
Project Manager 1	Wed 10/13/10	P	Work				\$150.00/hr	\$200.00/hr
Project Manager 2	Tue 10/26/10	P	Work				\$150.00/hr	\$200.00/hr
System Analyst 1	Thu 3/10/11	S	Work				\$75.00/hr	\$95.00/hr
System Analyst 2	Tue 3/1/11	S	Work				\$75.00/hr	\$95.00/hr
System Analyst 3	Mon 3/7/11	S	Work				\$75.00/hr	\$95.00/hr
Developer 1	Tue 4/26/11	D	Work				\$60.00/hr	\$80.00/hr
Developer 2	Thu 5/12/11	D	Work				\$60.00/hr	\$80.00/hr
Developer 3	Mon 5/30/11	D	Work				\$60.00/hr	\$80.00/hr
Laptop1	Wed 10/13/10	L	Material				\$200.00	
Laptop2	NA	L	Material				\$200.00	
Laptop3	NA	L	Material				\$200.00	
Laptop4	Wed 6/8/11	L	Material				\$200.00	
Laptop5	NA	L	Material				\$200.00	
PDA1	Wed 10/13/10	P	Material				\$75.00	
PDA2	NA	P	Material				\$75.00	
Cellphone1	Wed 10/13/10	C	Material				\$50.00	
Cellphone2	NA	C	Material				\$50.00	
Cellphone3	NA	C	Material				\$50.00	
Cellphone4	NA	C	Material				\$50.00	
Cellphone5	NA	C	Material				\$50.00	
Desktop1	Wed 10/13/10	D	Material				\$100.00	
Desktop2	NA	D	Material				\$100.00	
Desktop3	NA	D	Material				\$100.00	

Level Now

☒ Entire pool

☐ Selected resources

OK Cancel

Microsoft Project - Exercise1

File Edit View Insert Format Tools Project Report Collaborate Window Help

Type a question for help

New Resource From

Tasks Resources Track Report

	Resource Name	Start	Initials	Type	Email Address	Windows User Account	Group	Standard Rate	Overtime Rate
1	Project Manager 1	Wed 10/13/10	P	Work				\$150.00/hr	\$200.00/hr
2	Project Manager 2	Tue 10/26/10	P	Work				\$150.00/hr	\$200.00/hr
3	System Analyst 1	Thu 3/10/11	S	Work				\$75.00/hr	\$95.00/hr
4	System Analyst 2	Tue 3/1/11	S	Work				\$75.00/hr	\$95.00/hr
5	System Analyst 3	Mon 3/7/11	S	Work				\$75.00/hr	\$95.00/hr
6	Developer 1	Tue 4/26/11	D	Work				\$60.00/hr	\$80.00/hr
7	Developer 2	Thu 5/12/11	D	Work				\$60.00/hr	\$80.00/hr
8	Developer 3	Mon 5/30/11	D	Work				\$60.00/hr	\$80.00/hr
9	Laptop1	Wed 10/13/10	L	Material				\$200.00	
10	Laptop2	NA	L	Material				\$200.00	
11	Laptop3	NA	L	Material				\$200.00	
12	Laptop4	Wed 6/8/11	L	Material				\$200.00	
13	Laptop5	NA	L	Material				\$200.00	
14	PDA1	Wed 10/13/10	P	Material				\$75.00	
15	PDA2	NA	P	Material				\$75.00	
16	Cellphone1	Wed 10/13/10	C	Material				\$50.00	
17	Cellphone2	NA	C	Material				\$50.00	
18	Cellphone3	NA	C	Material				\$50.00	
19	Cellphone4	NA	C	Material				\$50.00	
20	Cellphone5	NA	C	Material				\$50.00	
21	Desktop1	Wed 10/13/10	D	Material				\$100.00	
22	Desktop2	NA	D	Material				\$100.00	
23	Desktop3	NA	D	Material				\$100.00	

Level Now

☐ Entire pool
☒ Selected resources

OK Cancel

start Uskov Office_2007_Project... CS591_Lab01_Manu... Microsoft Project - E... CS591_Lab01_Manu... 10:48 AM

Microsoft Project - Exercise1

File Edit View Insert Format Tools Project Report Collaborate Window Help

Type a question for help

New Resource From

Tasks Resources Track Report

Project Manager 1

Specify Resources

Do you want to:

☐ Add resources from Project Server
☐ Add resources from the company Address Book
☐ Add resources from the company directory
☐ Enter resources manually

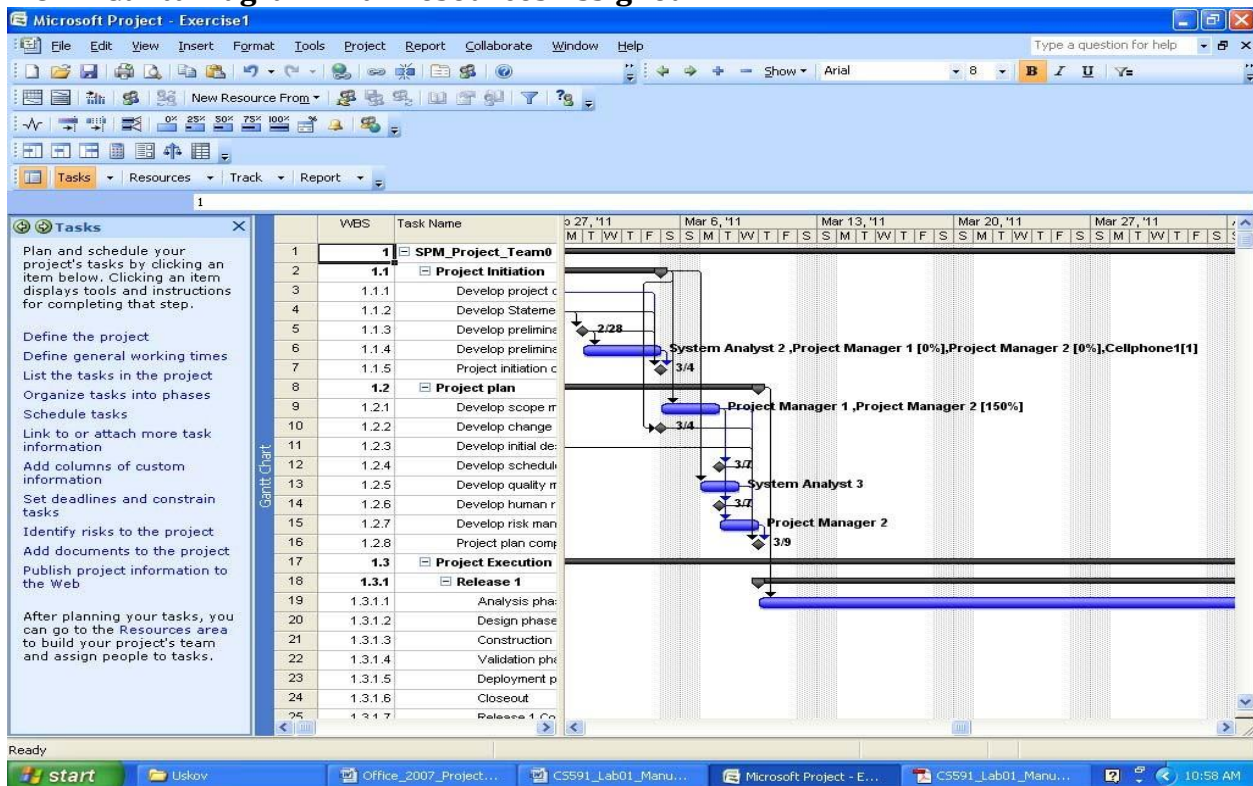
Done

	Resource Name	Start	Initials	Type	Email Address	Windows User Account	Group	Standard Rate	Overtime Rate
1	Project Manager 1	Wed 10/13/10	P	Work				\$150.00/hr	\$200.00/hr
2	Project Manager 2	Wed 10/13/10	P	Work				\$150.00/hr	\$200.00/hr
3	System Analyst 1	Thu 3/10/11	S	Work				\$75.00/hr	\$95.00/hr
4	System Analyst 2	Tue 3/1/11	S	Work				\$75.00/hr	\$95.00/hr
5	System Analyst 3	Mon 3/7/11	S	Work				\$75.00/hr	\$95.00/hr
6	Developer 1	Tue 4/26/11	D	Work				\$60.00/hr	\$80.00/hr
7	Developer 2	Thu 5/12/11	D	Work				\$60.00/hr	\$80.00/hr
8	Developer 3	Mon 5/30/11	D	Work				\$60.00/hr	\$80.00/hr
9	Laptop1	Wed 10/13/10	L	Material				\$200.00	
10	Laptop2	NA	L	Material				\$200.00	
11	Laptop3	NA	L	Material				\$200.00	
12	Laptop4	Wed 6/8/11	L	Material				\$200.00	
13	Laptop5	NA	L	Material				\$200.00	
14	PDA1	Wed 10/13/10	P	Material				\$75.00	
15	PDA2	NA	P	Material				\$75.00	
16	Cellphone1	Wed 10/13/10	C	Material				\$50.00	
17	Cellphone2	NA	C	Material				\$50.00	
18	Cellphone3	NA	C	Material				\$50.00	
19	Cellphone4	NA	C	Material				\$50.00	
20	Cellphone5	NA	C	Material				\$50.00	
21	Desktop1	Wed 10/13/10	D	Material				\$100.00	
22	Desktop2	NA	D	Material				\$100.00	
23	Desktop3	NA	D	Material				\$100.00	

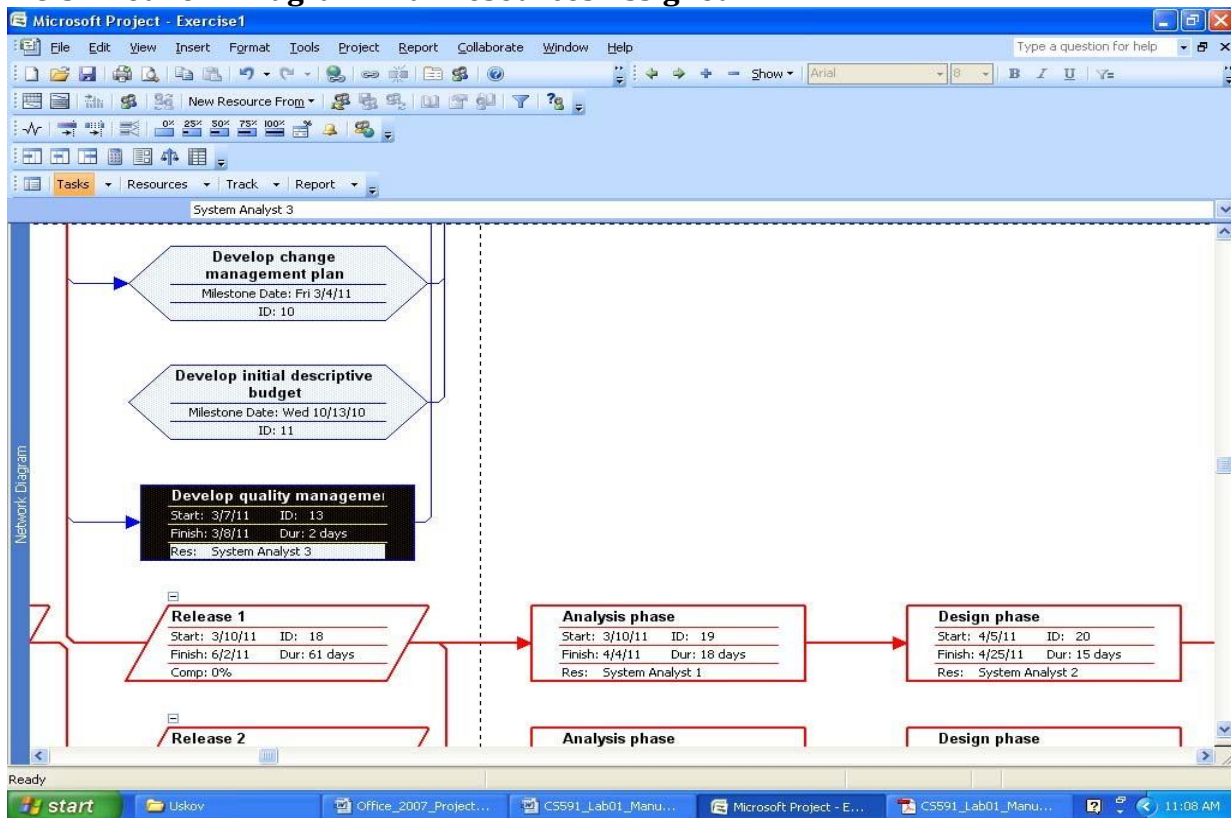
Ready

start Uskov Office_2007_Project... CS591_Lab01_Manu... Microsoft Project - E... CS591_Lab01_Manu... 10:53 AM

2.8.2 Gantt Diagram with Resources Assigned



2.8.3 Network Diagram with Resources Assigned



2.8.4 Assigning Resource Availability

The screenshot displays the Microsoft Project - Exercise1 interface. The Resource Information dialog box is open, showing the Resource Availability tab. The resource 'System Analyst 1' is selected, with a booking type of 'Committed' and a type of 'Work'. The Resource Availability tab shows a calendar for October 2010, with a date range from Monday, 10/11/10 to Friday, 10/15/10. The calendar shows the resource is available for 100% of the time.

Resource Information Dialog Box:

- General:** Resource name: System Analyst 1, Initials: S, Email: Windows Account..., Booking type: Committed, Type: Work.
- Costs:** (Empty)
- Notes:** (Empty)
- Custom Fields:** (Empty)
- Resource Availability:** Available From: Mon 10/11/10, Available To: NA, Units: 100%.

Resource Usage Table:

Resource Name	Work
Develop prelimi	0 hrs
Develop scope	8 hrs
Project Closeout	0 hrs
Project Manager 2	
Develop prelimi	
Develop scope	
Develop change	
Develop initial c	
Develop schedu	
Develop human	
Develop risk ma	
Project Closeout	
System Analyst 1	
Analysis phase	
Closeout	
System Analyst 2	
Analysis phase	
Develop prelimi	
Closeout	
Analysis phase	
Design phase	
Closeout	
Analysis phase	
Design phase	
Construction ph	
System Analyst 3	
Analysis phase	

Microsoft Project - Exercise1

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25% 50% 75% 100%

Tasks Resources Track Report

Specify people and equipment for the project

Specify the booking types for resources

Define working times for resources

Assign people and equipment to tasks

Link to or attach more resource information

Add columns of custom information

Publish project information to the Web

Build your project's team by assigning people to tasks by clicking an item below. Clicking an item displays instructions for completing that step.

Specify people and equipment for the project

Specify the booking types for resources

Define working times for resources

Assign people and equipment to tasks

Link to or attach more resource information

Add columns of custom information

Publish project information to the Web

After entering task and resource information, you can go to the Track area to track and manage your project as it progresses.

Resource Usage

Work	Details	M	T	W	T	F	S	S	M
0 hrs	Work								
8 hrs	Work								
0 hrs	Work								
16 hrs	Work								
0 hrs	Work								
0 hrs	Work								
0 hrs	Work								
0 hrs	Work								
0 hrs	Work								
16 hrs	Work								
0 hrs	Work								
176 hrs	Work								
144 hrs	Work								
8 hrs	Work								
24 hrs	Work								
264 hrs	Work								
32 hrs	Work								
120 hrs	Work								
8 hrs	Work								
24 hrs	Work								
56 hrs	Work								
24 hrs	Work								
64 hrs	Work								
16 hrs	Work								

Ready

Microsoft Project - Exercise1

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Tasks Resources Track Report

System Analyst 1

Resource Name	Start	Initials	Type	Email Address	Windows User Account	Group	Standard Rate	Overtime Rate
Project Manager 1	Wed 10/13/10	P	Work				\$150.00/hr	\$200.00/hr
Project Manager 2	Tue 10/26/10	P	Work				\$150.00/hr	\$200.00/hr
System Analyst 1	Thu 3/10/11	S	Work				\$75.00/hr	\$95.00/hr
System Analyst 2	Tue 3/1/11	S	Work				\$75.00/hr	\$95.00/hr
System Analyst 3	Mon 3/7/11	S	Work				\$75.00/hr	\$95.00/hr
Developer 1	Tue 4/26/11	D	Work				\$60.00/hr	\$80.00/hr
Developer 2	Thu 5/12/11	D	Work				\$60.00/hr	\$80.00/hr
Developer 3	Mon 5/30/11	D	Work				\$60.00/hr	\$80.00/hr
Laptop1	Wed 10/13/10	L	Material				\$200.00	
Laptop2	NA	L	Material				\$200.00	
Laptop3	NA	L	Material				\$200.00	
Laptop4	Wed 6/8/11	L	Material				\$200.00	
Laptop5	NA	L	Material				\$200.00	
PDA1	Wed 10/13/10	P	Material				\$75.00	
PDA2	NA	P	Material				\$75.00	
Cellphone1	Wed 10/13/10	C	Material				\$50.00	
Cellphone2	NA	C	Material				\$50.00	
Cellphone3	NA	C	Material				\$50.00	
Cellphone4	NA	C	Material				\$50.00	
Cellphone5	NA	C	Material				\$50.00	
Desktop1	Wed 10/13/10	D	Material				\$100.00	
Desktop2	NA	D	Material				\$100.00	
Desktop3	NA	D	Material				\$100.00	

Ready

Setting-up Resources

Statement Purpose:

The main purpose of conducting this LAB is to give an idea about the working of MS Project specially related to Resources.

Resources:

MS Project works with three (3) types of resources: Work, material and cost. Work Resources are the People and Equipment that do the work of the project

Individual People can be identified by their Name (like Mr. Ahmad) or designation (like Manager) and group of people resources can be defined by their common skills like programmers.

There are two types of work resources in MS Project. One is people and the other one is equipment.

People resources might be flexible in tasks they can perform (more than one jobs), however, equipment resources tend to be more Specialized. So be careful in assigning both the resources.